



Aerial firefighter radio communication performance in a virtual training system: radio communication disruptions simulated in VR for *Air Attack Supervision*

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Abstract

Communication disruptions are frequent in aerial firefighting. Information is more easily lost over multiple radio channels, busy with simultaneous conversations. Such a high bandwidth of information throughput creates mental overload. Further problems with hardware or radio signals being disrupted over long distances or by mountainous terrain make it difficult to coordinate firefighting efforts. This creates stressful conditions and requires certain expertise to manage effectively. An experiment was conducted which tested the effects of disrupting users communications equipment and measured their stress levels as well as communication performance. This research investigated how realistic communication disruptions have an effect on behavioural changes in communication frequency, as well as physiological stress by means of measuring heart rate variability (HRV). Broken radio transmissions created a greater degree of stress than background chatter alone. Experts could maintain a more stable HRV during disruptions than novices, which was calculated on the change in HRV during the experiment. From this, we deduce that experts have a better ability to manage stress. We also noted strategies employed by experts such as relaying to overcome the radio challenges, as opposed to novices who would not find a solution, effectively giving up.

Keywords Virtual reality · Simulation · Aviation · Radio communications · Firefighting · CRM

1 Introduction

This paper reports on research that simulates radio communication training within a virtual training environment for aerial firefighting. The focus of this research is on the evaluation of an immersive virtual reality (VR) training system, which reproduces radio communication problems often experienced in normal aerial firefighting conditions.

In this research, our aim is to create a realistic virtual training environment for wildfire aerial firefighting, by simulating radio challenges that are commonplace in this occupation, so that it can be explored as a variable for creating better training practices and learning opportunities in VR training.

VR simulation is shown to be effective for training in complex high-stress environments such as aviation [12,16,34],

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Fig. 1 (Left) A field exercise in the air operations platform (AOP) helicopter. (Right) The multi-sensory AOP simulator

firefighting [7], police [25], disaster, and military applications [3], to create training scenarios for practising stress management [38], situation awareness (SA) [6], and naturalistic decision-making (NDM) [10].

Traditional aviation-based firefighter training involves expensive, high-risk live aerial field exercises. Air attack supervisors (AAS) in training become airborne in a typical air operations platform (AOP) helicopter over a countryside terrain (Fig. 1, Left). When the AOP is in position over the fireground, other fixed-wing aircraft and helicopters proceed to drop water between a set of flags representing the fire. AAS trainees would observe the other aircraft over the fireground, providing feedback about their height, delivery, and approach vector. AAS would also assign additional tasks by providing instructions to the pilots and act as an interface to ground crew and operations management. To recreate fire behaviour, the flags are moved periodically. The simulated fireground quickly becomes a busy and sometimes chaotic space, with many different people and vehicles operating in the same area, creating a potential for incidents. In addition to these risks, it is difficult to control environmental aspects to create repeatable training exercises.

These challenges make VR simulations for AAS (Fig. 1 Right) a feasible option for safe and repeatable situated task-based training in addition to existing field exercises. Virtual reality firefighter training can provide complex and varied scenarios with greater control and flexibility over environmental conditions [7]. Virtual training often provides means for making minor changes to the conditions or parameters relevant to the occupation, which in this case is radio communication challenges. Based on previous research, situated virtual training with minor variances to the tasks could lead to better skill transference [9].

Effective radio communication plays a pivotal role in many industries, particularly in emergency domains [35]. It is a key aspect of crew resource management (CRM) and provides shared SA ensuring the effectiveness and safety of emergency responders [18]. Domains such as firefighting [35], the military [11], aviation [27], and air traffic

control (ATC) [19] all depend heavily on radio communication. Missed or broken radio messages have caused fatalities within these domains [35], directly causing harm, danger, and catastrophic situations which could have been avoided by practicing more effective communication [27,35]. Tasks in these domains are often high in complexity, which makes performance dependent on SA [21]; therefore, natural, effortless communication is a very important factor to effective training. Many team-based emergency training simulation scenarios such as aerial firefighting rely upon radio communication to share SA and deliver orders to other aircraft pilots and ground crew [35].

In aerial firefighting, CRM is maintained over radio communication, in contrast to other crisis management situations where supervisors could have access to computerized decision support systems [36]. The AAS is expected to manage a large throughput of complex information, holding many conversations with different people simultaneously. This causes mental stress [28] known to AAS as *helmet fire* due to the use of the flight helmet. AAS must maintain a constant communication with the aircrew to manage the aerial operation and keep the operations manager on the ground up to date via radio. Any unexpected events in a wildfire can cause the AAS additional stress which can be compounded in such a NDM environment. In a virtual training system, we can artificially create these unexpected situations to create task-based stress. Radio communication is an effective tool that is relied upon in emergency response domain as well as in aviation; therefore, it is also a critical component to the simulation.

This study is the first research that evaluates communication problems in virtual training environments in the domain of aerial firefighting, specifically AAS. This field is of particular interest, and this research provides insight into how these professionals operate and how we can create appropriate technology to support them. The immersive system exposes trainee AAS to complex wildfire situations. Additional mentally stressful effects such as *helmet fire* (i.e. information overload) and other communication disruptions can be simulated. This enables effective practice for aerial firefighting

using typical radio communication challenges. The multi-person situated training environment provides repeatable task-focused training exercises in a safe, multi-sensory virtual simulation.

Observing radio communication behaviour in this virtual laboratory environment will help us to understand the effects of occupational stressors as well as the problems of trainees. We can discover ways to improve the decision-making quality, communications skill, and overall confidence of AAS helicopter copilots. In VR, many specific training scenarios and situations can be designed for offering an array of variables that are commonplace in this complex domain. General problem areas such as radio communication protocol can be focused on as a team, and specific problem areas for each individual trainee can be targeted, making this a valuable contribution. Our ultimate goal is to fully replicate the AAS environment in its entirety, creating a multi-sensory virtual *Air Attack* simulator with occupationally derived situations, tasks, and challenges.

2 Background

In this section, we explore the existing literature around communication factors and errors in different domains of expertise. We also discuss the effects of disruption to radio transmissions and environmental noise in the cockpit. Additionally, we describe simulation training in general and within the aviation industry. Lastly, we discuss factors of stress and physiology.

Radio communication is an essential component of aviation [27]. It is frequently used to coordinate air space occupation in order to avoid collisions, to negotiate the order in which aircraft takes off and land, and to communicate with ATC personnel. ATC is responsible for the large-scale coordination of air traffic.

The AAS works as the copilot in the AOP helicopter, performing decision-making and ATC-like tasks, providing global SA [6] over the fire ground for the operations and intelligence teams as an airborne observation platform. The AAS is required to listen carefully to emergency radio traffic and talk to many people using a multi-channel radio communications system. Other essential emergency radio chatter over these channels is also frequent, with other aircraft pilots and ground crews holding multiple conversations that the AAS may need to pay attention to. This high-bandwidth mental demand, combined with the long-term environmental exposure caused by sitting for extensive, repeated duration's in a helicopter over a raging fire ground, creates a high degree of stress.

Radio communications are the primary interface for the AAS, so we consider what effects that systematic problems with radio communications have on AAS stress levels, as a

means to provide stress resilience training to AAS, which has been found to increase proficiency in stressful flight environments [16].

2.1 Disruptions to message transmission

Equipment failure and broken transmissions can occur frequently as people or vehicles move in and out of rough terrain [11], which breaks the radio signals and causes further disruption to the communication, by being inaudible, misunderstood, or not received at all [35]. These factors need to be considered when developing a VR training system for team-centric occupations spanning over large distances, where communication is vital and problems exist.

2.2 Radio communication in the military

In a survey of Finnish military pilots, problems with communications occurred an average of fourteen per cent of the flight time [11]. These disruptions included multiple speakers talking on the same radio channel causing overlapping speech and missing acknowledgments, high levels of background noise, especially during helicopter operations, and technical problems with the hardware. The two main contributing factors of radio communications are human errors in communication such as not following correct radio protocol and equipment or mechanical failure [11,27].

2.3 Radio communication in fire and emergency

Varone [35] found that operating on limited radio channels was inadequate for dispatchers to perform all the functions required during a fire event. These tasks include dispatching apparatus, fire ground operations, recall of off-duty personnel, and emergency medical calls. He cited many occasions where dispatcher's would override a radio transmission of crew who were present on the fire ground, including messages that were requesting urgent assistance from trapped firefighters. Radio communications can easily become flooded with a high volume of radio traffic at any given time during a fire event [11]. This disruption prevents messages from getting across, creating potentially dangerous situations.

2.4 Aircraft environment noise

Aircraft noise is characterized by short periods of high levels of noise and in the case of helicopters can fluctuate suddenly when changing altitude or performing turning manoeuvres. The sound from the helicopter engine and rotor blades reaches levels of between 90 and 110 dB, depending on the number of rotors [33]. Although pilots often use noise cancelling headsets, the vibrations from the engine, rotor blades, and turbulence can still be felt through the aircraft

chassis and the seats. In the case of aerial wildfire firefighting, it is normal to be subjected to these effects over a prolonged exposure. Without sufficient rest, this may lead to potential physical or mental health issues [17] such as changes in emotional state, changes in behaviour (e.g. aggressiveness, being withdrawn), cognitive issues (e.g. concentration, indecisiveness), and physical effects (e.g. nausea, heart rate, HRV).

2.5 Virtual simulation-based training

Virtual environments for training are an effective tool for transferring skills to real-work environments and can even be comparable with the training effects of real-world exercises for collaborative tasks [2]. By incorporating communication failures, a team-based collaborative training system can achieve a higher fidelity by mimicking the actual task more appropriately, which results in better skill transference to real-world situations [32].

2.6 Simulation training in aviation and ATC

Simulations are used extensively for aviation training [12] and for ATC to reduce risk and increase training effectiveness. Communication is often overlooked in simulation but is crucial to the occupation [12]. Fidelity of simulation training can be a low, medium, to a high level of real-world accuracy [34]. Future recommendations for simulation training in this role include voice recognition and synthesis to replace pseudo pilots and enable training at home by using networking and teleconferencing technologies in a location-independent or virtual manner [34].

2.7 Physiological and psychological responses to stress

Communication disturbances directly relate to the decrease in SA of the AAS by impairing the complete assessment of the situation. This loss in SA can increase the mental loading experienced [25]. Furthermore, working back-to-back shifts without adequate sleep further adds to the increased stress levels experienced by emergency personnel [24]. By exposing candidates to stressful situations, in this case, loss of communication, it can be utilized to habituate the handling of probable communication failures. This type of stress inoculation has shown promise in simulations of high-risk tasks to prevent adverse psychological reactions during actual performance [1,38]. When situational demands outweigh the coping capabilities of the individual, stress symptoms are inevitable [8]. Training to cope with these demands in a VR simulation can be used to prevent this negative effect from occurring.

2.8 Measuring physiological stress

Wearable physiological monitors are appropriate tools for measuring physiological stress in simulated and real firefighting activities [20,30] as they can record heart rate variability (HRV) and breathing rate (BR) in a relatively non-intrusive way, in a virtual training environment [37]. This provides an effective method to capture live physiological data without having to disrupt the flow of the virtual exercise [22].

2.9 Background research summary

Radio communication creates a high degree of cognitive demand for AAS, requiring them to handle a large bandwidth of information in a noisy environment. This critical component in team-based occupations, when managed well, can be the difference between success and failure. As effective as radio communication may be, it can also create mental stress when improper radio communication protocol is used or when equipment failure occurs. In the case of AAS, it can result in poor suppression efforts and catastrophic outcomes such as loss of life, making these situations stressful to pilots and decision makers. Simulated radio communication and the challenges around this have not been rigorously explored [12]; our research aims to rectify this.

3 Methodology

The effects of radio disruptions on the stress levels of trainees are an important consideration, since the tool is used heavily in both aviation and firefighting domain. The question is: How do more experienced trainees handle occupational stress? In this experiment, we expect novice users to feel more stress or to be unable to handle the situation as easily as a highly trained firefighter or the expert user group of AAS. A natural human response would be to show signs of heightened nervousness or physiological response such as changes in HRV or BR. However, we would expect that an expert would be less susceptible to the conditions reflected in their physiology, as they would be confident in their ability to perform up until the point where the task becomes overwhelmingly difficult. This would be reflected by their skill level or capacity to handle the situation in a typical noisy aircraft environment. When background radio chatter is present, a cocktail party [4] effect may take place where effects of helmet fire cause confusion or selective hearing. Experienced communicators would effectively manage multiple conversations simultaneously, ignoring conversations that do not pertain to them and keeping an ear on the other radio chatter. Additional miscommunication errors caused by equipment failure would cause further disruption.

It is uncertain what effect this cognitive workload of compounded communication stressors has on the physiological

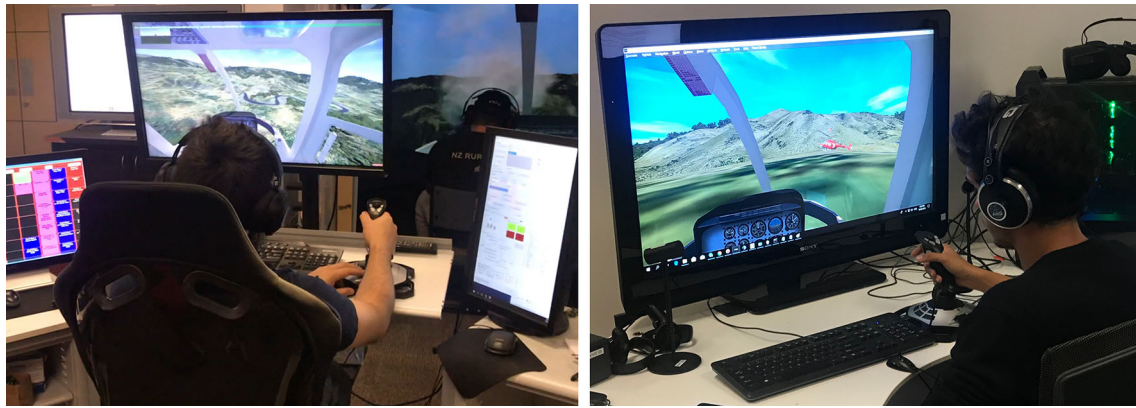


Fig. 2 Experimenter (Left) & Research assistant 1 (Right)

stress and performance ability of AAS. To understand the effects of communication disruption on the levels of stress and communication ability in AAS, we designed an experiment that involved three conditions. Communication and physiological data were captured in real time to show these effects, in combination with pre- and post-experiment questionnaires on stress and presence.

3.1 Hypotheses

Considering the effect and dependency of communications on the situated task, we looked at the common issues experienced with typical radio communication systems. We identified critical factors such as background radio chatter and signal disruptions as experimental conditions. A repeated action, a push to talk (PTT) foot switch, is used to enable radio communication. Based on this, the hypotheses for the experiment were as follows:

H1 Participants can effectively communicate and respond to human helicopter pilots, AI fixed-wing pilots, and the operations manager, using the simulated emergency radio communication system.

H1.1 There is a difference in the effectiveness of communication between participants with radio experience and participants without.

H2 Participants have a sense of presence in the virtual environment (VE) as measured with the I-Group Presence Questionnaire (IPQ).

H3 Added stressors in the system such as background chatter and disrupted communications have a noticeable effect on the physiological stress level of participants.

H3.1 There is an observable decrease in HRV.

H3.2 There is an observable increase in BR.

H4 The system will elicit an increase in perceived stress, measured by relevant subscales (energetic arousal, tense arousal,



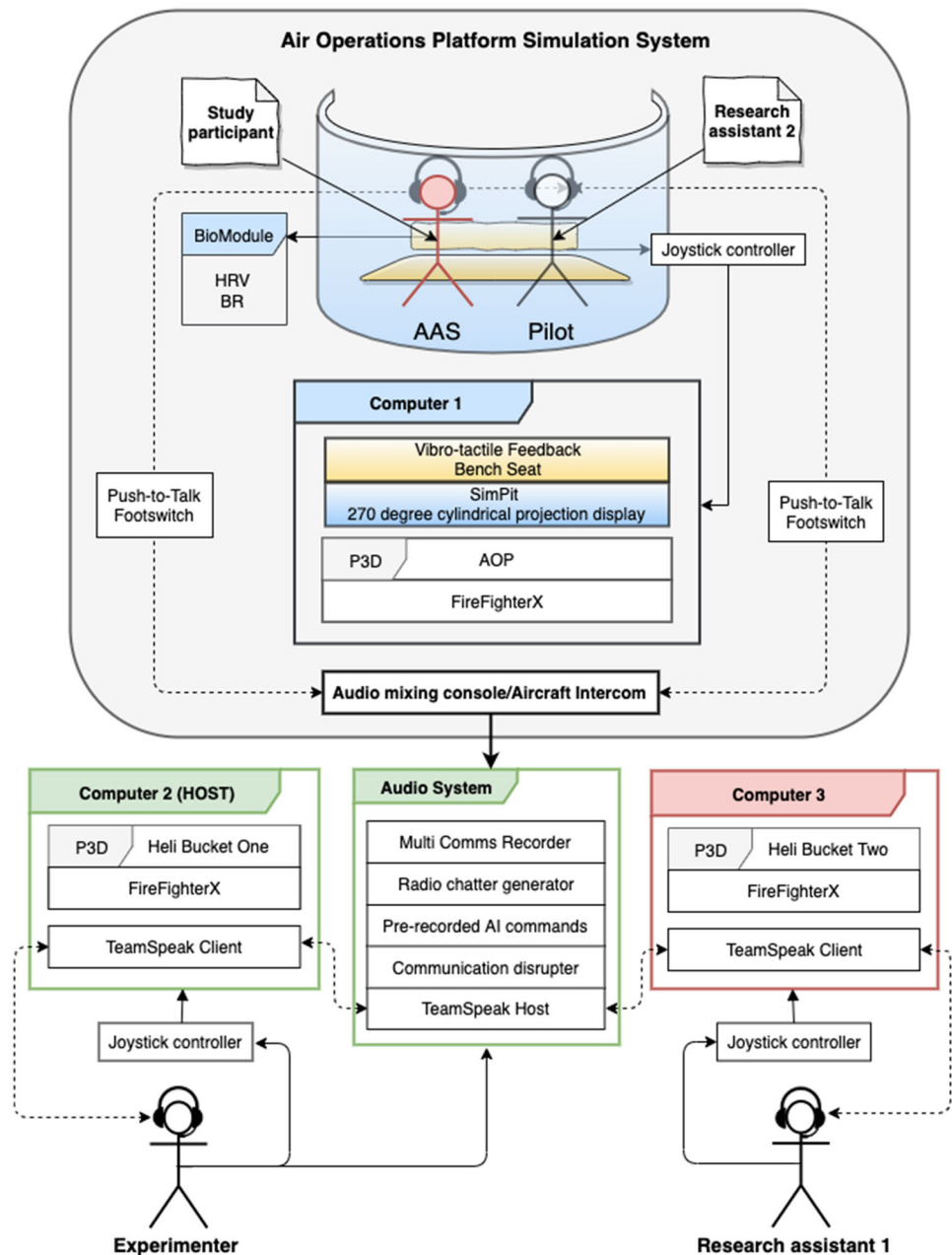
Fig. 3 Virtual air attack helicopter cockpit with a fire

anger/frustration, and concentration) of the Dundee Stress State Questionnaire (DSSQ).

3.2 Prototype AOP helicopter AAS training system

We combined a variety of technologies in our multi-sensory simulation environment including computer generated immersive visual stimulus, surround sound audio, and low-frequency vibro-tactile feedback. The system allowed a practicing AAS to coordinate with other aircraft pilots and follow the instruction of the operations manager in the simulation (see Fig. 2 Left & Right) to attack the fire by giving verbal instruction and requesting information using the provided communications interface. The air attack aircrew attempted to extinguish a virtual forest fire (Fig. 3). The team consisted of two real helicopter pilots and two computer-controlled AI fixed-wing aircrafts. The fixed-wing aircraft and the operations managers' verbal radio responses were triggered by the experimenter, using a pre-recorded message on a touch screen sound board. Figure 4 shows the complete system diagram for the experimental setup.

Fig. 4 Diagram of the experimental system, dotted lines show communication channel and solid lines physical interactions



3.3 Hardware

Three Intel Core i7-7700K CPUs at 4.20 GHz computer systems with 32GB DDR3 system memory, with a GTX 1080Ti graphics card, were used to run the simulation software. A Logitech Flight 3DPro Joystick was used as the main user interface (UI) on each of the computer systems to provide an intuitive interaction experience. Additionally, communication was provided with a headset to replicate the radio environment of the cockpit. A computer is needed to run the simulation software for each actor in the simulation. One computer was operated by the experimenter, another was

used by a research assistant who together covered the bucketing helicopter pilots, and a third computer was used for the AAS participant and a second research assistant, acting as the AOP pilot. The AOP was recreated visually inside a 270-degree cylindrical projection display (Fig. 5), a visually immersive display for the virtual environment, called the SimPit [29]. From the results of a previous experiment, the SimPit provided the most realistic operating environment, combining both real-world and virtual elements. It also produced less simulator sickness than the VR head-mounted display [5]. The visual display system was coupled with a vibro-tactile feedback chair, which consists of two Earth-



Fig. 5 Multi-sensory vibro-tactile virtual AOP cockpit. Pilot Research assistant 2 (Right) and AAS copilot (Left)

quake Sound Q10B tactile transducers [31]. This sensory stimulus gives the effect of rotor and engine vibration. A Zephyr BioModule [39] wireless heart rate ECG physiological monitor was used to record the HRV and BR data of study participants, operating as an AAS in the virtual environment. We maintained the temperature inside the SimPit to be approximately 25 °C. AI fixed-wing aircraft was launched by the host of the simulation but flew autonomously. To introduce and control different radio communication disruptions in the experiment, we connected the system to an audio mixer that could control the audio effects for the communications using faders and buttons. A touch screen monitor (Fig. 2 Left) was used to trigger pre-recorded responses from AI pilots and the operations manager using a digital soundboard UI.

3.4 Software

We used the flight simulator engine Prepar3d 4.1 [14] by Lockheed Martin (Figs. 3, 5), which enabled realistic flight behaviour, visual modelling of real-world locations, the environment and provided the network infrastructure to enable a collaborative multi-person environment. We also used the Lorby-SI FireFighterX plugin [13] which enabled fire effects, fire spread simulation, and fire extinguishing abilities which were used to simulate the wild fire event. It also provided the ability to extinguish the fires by dropping water from the aircraft. The digital soundboard for the fixed-wing aircraft AI and operations manager was created by pre-recorded phrases and then triggered by the experimenter (Fig. 2, Left) on the soundboard to give the impression that the AAS were communicating with real people, additionally to the two human helicopter bucket pilots, to create more helmet fire.

3.5 Experiment design

Participants were asked to perform radio communication tasks which required them to utilize the radio hardware to communicate with all parties in the simulation, the AOP pilot,

the two bucketing pilots, the AI fixed-wing support aircraft, and the operations manager.

The PTT foot switch was provided to communicate with those outside of the AOP, which is normal in such a helicopter radio system. The PTT foot switch activity is logged, so we could track the frequency of outgoing communication behaviour. To talk to the AOP pilot, participants could openly communicate using the on board intercom and use hand gestures to indicate their intention. All radio communication was recorded for further analysis.

To determine if the system induces stress, questionnaires (see Sect. 3.7.2) and a wireless physiology monitor were used.

3.6 Procedure

The conditions were presented in the following order, with no interruptions:

1. PTT radio, helicopter noise, and vibration feedback.
2. Condition one + background radio chatter.
3. Condition two + broken transmissions/signal failure.

The following process was carried out from the exact moment the participant arrived:

1. Welcome participant, provide information about the study and have the participant complete the consent form.
2. Activate and fit Zephyr Biomodule to participant to obtain a stable HRV (takes approx. 5 min).
3. Participant completes the demographics and pre-experiment questionnaires, while waiting for the HRV sensor to gain 100% confidence.
4. Participant performs breathing exercise for four minutes, when HRV is at 100% confidence to obtain the relaxed state HRV.
5. Participant now sits in virtual AOP copilot seat, ready to begin the training exercise.
6. Perform communications sound check, to ensure everyone can talk and for participant to become familiar with the PTT foot switch system.
7. Experiment scenario begins (condition 1), all helicopters to take off and fly towards fire ground.
8. 10 min in, introduce condition 2.
9. 15 min in, send in fixed-wing AI Aircraft 1.
10. 20 min in, introduce condition 3.
11. 25 min in, send in fixed-wing AI Aircraft 2.
12. 28 min low fuel warning; return to base.
13. 30 min in, landing at base.
14. Perform post-experiment breathing exercise.
15. Participant completes post-experiment questionnaire.
16. Remove Zephyr Biomodule from participant.

17. Thank participant for their time and reward them with a \$10 gift voucher.

3.7 Measures

In this section, we cover the types of measurements used in the experiment to deduce the effects on stress, the level of presence of the simulation, and whether or not our hypotheses were supported.

3.7.1 Validated questionnaires on presence and stress

Existing questionnaires are available for detecting presence and stress in a VR system. The I-Group Presence Questionnaire (IPQ) is a seven-point scale which can be used to detect presence in VR [26]. The Dundee Stress State Questionnaire (DSSQ) provides an effective way to profile stress within a system [15].

3.7.2 Physiological data

HRV and BR were measured. A lower HRV value or shorter inter-beat interval means the heart is beating more rapidly, indicating stress. The higher the BR, the more stress the user is experiencing. We obtained these data before, during, and after the stimulus exposure. To compare and interpret these data, we took the average of the HRV from each participant during the experiment. We took an average of 100 samples after the introduction of each stimuli and subtracted this from the peak HRV value reached during the initial breathing exercise. This was carried out due to individual physiological differences, so we consider this method unbiased towards people who have a generally high HRV.

We also compared the previous four-minute breathing exercise HRV versus the same post-breathing exercise (BE2) after the simulation experience to look for stress recovery, indicating whether participants were in fact subjected to stress due to an increase in HRV from the initial breathing exercise (BE1). A positive response here (greater than zero) indicates a better ability to recover from stress, implying the participant is better at handling stressful situations.

4 Results

A total of 25 participants were involved in the study, two females and 23 males. There were eight AAS, six firefighters without AAS experience but with radio experience, and eleven novices with no prior radio experience.

H1. Participants can communicate effectively with human and AI pilots and the operations manager using the emergency simulated radio communications system.

Table 1 Average number of foot switch presses during experimental phases between participants with and without radio experience

	No radio exp.	Radio exp.
(1) No disruption	27.50	21.77
(2) Background chatter only	27.83	24.54
(3) Chatter and signal failure	28.83	22.08

All participants could effectively use the system to communicate with all actors and provided visual feedback on the effectiveness of the aircraft in operation using the PTT foot switch, see Table 1.

H1.1 There is a difference in the effectiveness of communication between participants with and participants without radio communication experience.

Due to systematic errors during the experiment, some of the foot switch presses failed to be recorded for six of the participants. The remaining sample consisted of thirteen participants with and six participants without radio experience. Looking at the number of foot switch presses during the periods of (condition one) no communication disruption, (condition two) disruption through background chatter, and (condition three) disruption through background chatter plus occasional signal failure, we observed that participants with radio experience managed to more effectively communicate during the experiment (see Table 1). Needing less foot switch presses on average means that they were able to give their instructions more efficiently, making their overall communication more effective. Due to the rather small sample size and the additionally lost data, no further analysis was conducted.

H2. Participants feel presence within the virtual environment.

A one sample t test was used to evaluate the results of the IPQ survey, presented in Fig. 6. Generally, participants felt a strong sense of presence in the virtual environment. The combined score of the IPQ ($M = 4.5$, $SD = 0.70$) was significantly above the neutral midpoint (4.0) of the seven-point scale $t(24) = 3.549$, $p = 0.002$. The mean *general presence* score ($M = 5.72$, $SD = 1.021$) was significantly higher than the neutral midpoint, $t(24) = 8.420$, $p < 0.01$. The *spatial presence* mean ($M = 5.31$, $SD = 0.864$) was significantly greater than the neutral midpoint of 4.0, $t(24) = 7.590$, $p < 0.01$. The *involvement* mean was ($M = 4.60$, $SD = 1.53$) and was not found to be significantly higher than the midpoint ($t(24) = 1.956$, $p = 0.062$). *Experienced realism* mean was ($M = 3.10$, $SD = 0.510$) and was significantly lower than the neutral midpoint $t(24) = -8.818$, $p < 0.01$.

When questioned about the multi-sensory stimuli specifically, if the background vibration and helicopter noise was realistic: “engine sound and the vibrations make the simulation much more immersive”. It was remarked, however, that

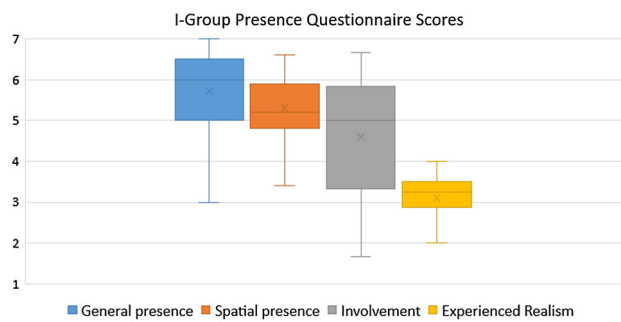


Fig. 6 Global IPQ scores with observable midpoint of 4

the vibration needs to be stronger when entering hover in the simulation.

H3. Added stressors in the system such as background chatter and disrupted communications have a noticeable effect on the physiological stress level of participants.

This hypothesis is answered by observing the results of the two sub-hypotheses **H3.1** and **H3.2**.

H3.1 Heart rate variability results.

We performed a repeated measures GLM data analysis and found that the average HRV differences from baseline were significantly different ($F(2, 48) = 19.10, p < 0.001$). The second phase ($M = -26.31, SD = 18.81, p < 0.001$) and the third phase ($M = -27.22, SD = 20.77, p = 0.001$) were significantly lower than the first phase ($M = -9.34, SD = 19.56$). The difference between the second and the third phases was not significant ($p = 0.941$). We also found a highly significant within-subject linear contrast for the first three phases, indicating that the HRV decreased with the progress of the phases.

H3.2 Breathing rate results

We performed a repeated measures GLM data analysis and found that the average BR differences were significantly different ($F(2, 48) = 9.84, p < 0.001$). The second ($M = 8.68, SD = 5.42, p < 0.003$) and third ($M = 8.56, SD = 5.35, p = 0.002$) were significantly lower than the first condition ($M = 10.78, SD = 4.59$). The difference between the second and the third conditions (Fig. 7) was not significant ($p = 0.995$).

We also found a highly significant within-subject linear contrast for the three phases, which indicates that the BR decreased with the phases. This suggests that participants reduced their respiratory frequency with the progress of the experiment.

H4. The use of the system elicited an increase in perceived stress, measured by relevant subscales (energetic arousal, tense arousal, anger/frustration, and concentration) of the DSSQ (Table 2).

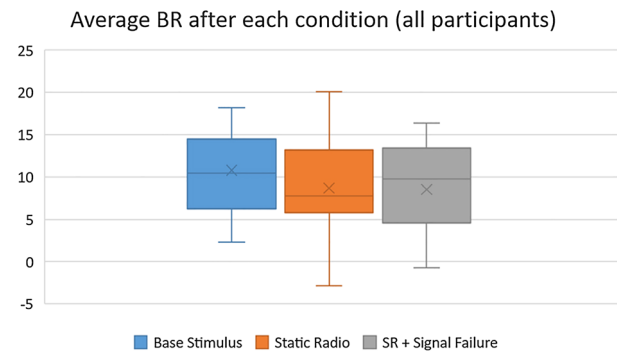


Fig. 7 Average BR for all participants in each condition

Table 2 Scores for DSSQ

	Before	After
Energetic arousal*	23.60 ± 3.20	26.48 ± 3.16
Tense arousal	14.60 ± 4.94	15.88 ± 4.94
Anger/frustration	18.44 ± 2.12	18.68 ± 2.38
Concentration*	17.24 ± 3.50	19.12 ± 3.00

*Statistical significance

Participant self-reported ratings of their experience during the use of the system were compared with their stress state before the intervention on relevant subscales of the DSSQ (Table 1). The repeated measures GLM statistical analysis showed that *energetic arousal* during exposure was significantly higher ($F(1, 24) = 18.805, p < 0.001$) and *concentration* ($F(1, 24) = 18.498, p < 0.001$). *Tense arousal* ($p = 0.222$) and *anger/frustration* ($p = 0.571$) were not significant.

In regard to the user experience of the stressors, 72% of participants (18/25) ranked condition three *distorted communications* as the most stressful, the majority of whom reported that it made them feel like they were not in control of the situation. In this condition, participants stated that they had to rely on other pilots to pass along messages, clarify messages multiple times, and were uncertain concerning the instructions given and status of it being received. Two participants with previous radio experience described the distorted communications as not stressful at all as this is “standard for anyone having radio experience”. The background chatter was stated to be quiet enough to be tuned out by several participants who possessed radio experience.

5 Discussion

All twenty-five participants communicated to human and AI Pilots using the foot switch communication interface. All participants effectively communicated a tactical vantage point to the AOP pilot. Our system was proved to be usable by

trainees of all experience levels as they all completed the task effectively, producing the number of foot switch presses for each user group as seen in Table 1. From this information, we conclude that **H1** is supported, as all participants successfully grasped the concept of using the foot switch to communicate outside of the AOP, while being able to communicate effectively locally within the AOP.

Regarding **H1.1**, the limited amount of available data makes interpretation of the results a difficult task, while we can see that overall the amount of foot switch presses is higher for the participants without radio experience as shown in Table 1. With the current limitation in sample size, we cannot conclude that our **H1.1** sub-hypothesis is supported by the data. Further research needs to be conducted, with a larger pool of participants to make any solid conclusions about **H1.1**. Considering **H2**, participants felt a significantly different score in both generally presence and spatially presence within our system, supporting this hypothesis. Involvement was not significantly different from the observed midpoint score. The experienced realism was significantly lower than the observable midpoint score.

As outlined by Schubert et al. [26], the concepts of involvement and spatial presence are distinct and independent during measurement. The significantly high spatial presence measures could be explained by the high fidelity of the simulated cockpit during the experiment. Using the same motions in the virtual environment as in the actual task has an influence on the experience of spatial presence, while not influencing experienced realism and involvement to the same degree [23]. It could be argued that the heavy focus on audio stimuli and communication took away some attention from the virtual environment during the experiment, which could have resulted in the lower involvement scores. This could also explain the low experienced realism scores. Nevertheless, the general presence score insinuates that participants did feel *as if they were there*. A heavier load on the spatial presence scores was found when obtaining the score for general presence, which comprised of all three factors. With the general presence score being significantly higher, we conclude the **H2** is at least partially supported.

The reduction in HRV (less time between heart beat intervals) from condition one to condition three suggests that the stress level of participants increased with the addition of stressors (see Fig. 8). We can see that the overall recovery rate of HRV from the experiment is increasing, which indicates that the experiment induced stress (HRV Recovery BE2, Fig. 8). We also observed that the HRV during condition two (background radio chatter) and condition three (background radio chatter plus distorted communications) are significantly lower than condition one. But they both show differences between the initial breathing exercise and the base stimulus, partially supporting our sub-hypothesis **H3.1**. We were expecting to see a significant drop in HRV considering

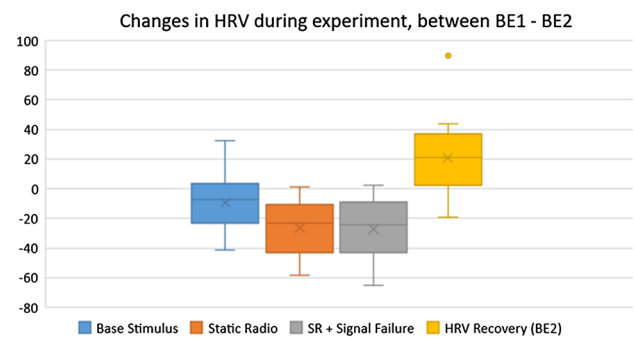


Fig. 8 Average HRV means after each condition compared with the HRV at starting breathing exercise (BE1)

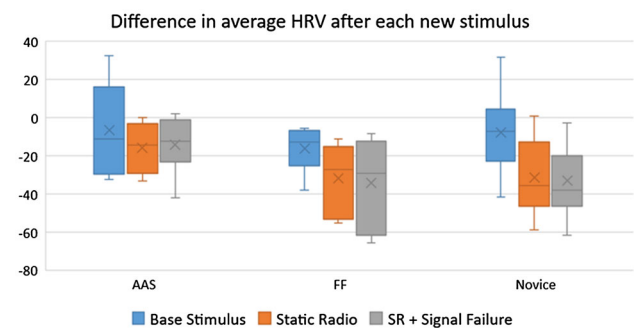


Fig. 9 Average HRV of each expertise, across each condition

the effects were layered, from condition two (background radio chatter) to condition three (broken transmission/signal failure), which was still true when looking within the individual groups of expertise. However, since no significant reduction in HRV occurred overall, **H3.1** is only partially supported.

We observed that the expert user group, the AAS group, were better at maintaining their HRV than both the general firefighter group and the novice group (Fig. 9). We expected an increase in BR with the addition of stressors as this would be linked with more stress perceived by the participants'. However, the results (see Fig. 7) show the BR decreased between conditions, which could also be an effect of the participants' reduction in talking frequency. The duration of the experiment may have led to fatigue effects, reducing the desire to communicate. This suggests that BR may not be an effective measure due to the influence talking has on BR. In the case of **H3.2**, it is not supported. From the results of the two physiological measurements HRV and BR, we conclude that **H3** is partially supported.

Considering **H4**, the DSSQ scores show an overall increase in energetic arousal, suggesting that the system was engaging. There was also a reduction in hedonic tone, but this was not significant. There was an increase in success motivation, primarily from the AAS participants, and an increase to intrinsic motivation, which was primarily in the novice group. Overall motivation shows signs of improvement after the

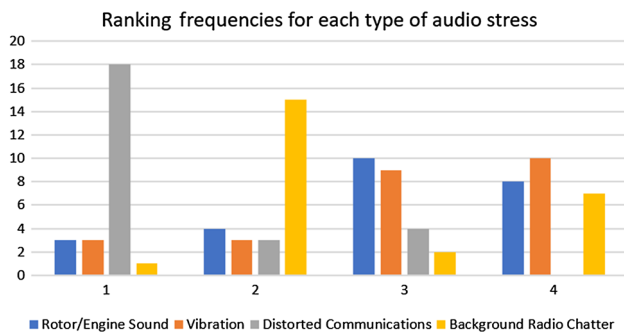


Fig. 10 Rankings of audio stressors (1 is least favoured)

experiment. There was an overall reduction in self-focused attention, suggesting participants were thinking more about others, perhaps feeling responsible for their well-being. Concentration generally increased for each group, where the control and confidence scores dropped in value, which suggests that the system made participants more aware of the difficulties involved in radio communication. We conclude that H4 is partially supported. The distorted communications were ranked to be the least favoured stressor by our participants (Fig. 10), which is supported by the measured HRV, suggesting that it was the most stressful as experienced physiologically.

Experienced AAS reported that the distorted communications were realistic of situations that can occur in the field. Experienced AAS found that when the communications system was broken, they would adapt their strategy and come up with techniques to relay the information such as using one of other helicopter pilots to pass messages across to the others to ensure their message gets across. Using tactics like this shows a higher level of experience and the ability to adapt to the conditions, whereas novices would tend to give up on the transmission entirely. Some participant responses to open ended questions about the system and the stressors are as follows: “With distorted communications and background radio chatter, you’re fearing to miss teammates communications and wing craft interventions. But the engine sound and the vibrations make the simulation much more immersive”. “It mean(t) I had to ask them (the pilots) to repeat what they were saying and took my mind off the fire, which made it hard to make choices. The background radio chatter was hard because you were trying to listen to the helicopters, but other people were talking”. “When the communication became difficult to hear I did not feel in control of the incident”.

The background radio chatter was mostly ignored according to participants, suggesting that a cocktail party effect had taken place [4], “Background radio chatter was fine as long as it didn’t interfere with the communications”, “I ignored most of radio noise/chatter as ambient sound”. This is an interesting result as we expected the background chatter to have an effect on the participants’ communication ability.

However, since the background chatter was somewhat garbled, incomprehensible, or lacked information related to the scenario we were replicating, it may have made the chatter easier to ignore, producing an undesired effect. With this in mind, a more sophisticated background chatter which triggers mental stress could be designed where conversations between firefighters facing difficulty on the fireground are held, making frequent radio calls and speaking erratically, causing radio disruption.

6 Conclusion and future work

As we have learned from subject matter experts, many situations that arise in the firefighting domain are often caused by human behaviour and error. For example, lack of radio communications between the pilots can mean that there is actually a problem unfolding, such as a drop in the confidence of pilots, radio systems malfunctioning, or that the pilots are experiencing fatigue from flying over long and intense periods of time. Pilots may also develop or possess an ego, especially when competing helicopter companies are working together on a fire. Human traits like these can lead to potential problems and make it difficult for AAS to make decisions about the pilots in their crew, to keep them flying, or to stand them down. These examples and others like it are worth investigating in situated training exercises and should be included in future training systems for a psychologically immersive experience.

We have shown that our system works effectively and provides an accurate AAS radio environment. A range of participants from those who had AAS experience, general firefighters and the novice group, all successfully completed the task, in each of the conditions. The distorted communications or signal failure effect (condition three) was the most effective at creating stress, and the background radio chatter (condition two) was not as difficult to handle for participants, but was considered to be second most stressful by participants and produced a similar level of HRV. BR was not found to be a good measurement for stress in this case, as the frequent need to talk has an effect on the BR, which tended to deteriorate over time.

Many participants felt that the background chatter needed to be more related towards the exercise for it to be worth paying attention to. In real life, the background chatter experienced on emergency radio generally contains vital information and it would be detrimental to train people to ignore it. By injecting more relevant radio events pertaining to the AAS or to the operation as a whole, it would increase the amount of potentially relevant information that needs to be taken into consideration or ignored. It would also benefit from being slightly more dynamic or randomized, so that no two radio calls are the same or that it reacts to the

user speaking or tries to interrupt them. This would create a greater cognitive workload which better simulates the AAS radio environment. Additionally, the wireless HRV sensor could be used to dynamically change the frequency or intensity of the background chatter based on the user stress level, creating a more reactive training system.

We plan to improve the realism of the communication system and make it more intelligent, so that it reacts to different terrains and topologies, for example, making the radio distortion effects depend on line of sight, the distance users are away from each other, or if there are any obstacles blocking the transmission. Another consideration to the communication system would be to design a replica console from a typical AOP helicopter radio and gauge cluster. This would recreate the physical interaction experienced, creating muscle memory and additional mental workload, which would be another opportunity to study the effects of the combined mental and physical stressors between different levels of expertise.

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